A Systematic Mapping of Large Language Models for Improving Natural Language Software Requirements

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Abstract: The advancement of Large Language Models (LLMs) has impacted Requirements Engineering, an area traditionally challenged by the evaluation of the quality of requirements expressed in natural language. Despite the growing interest from the scientific community, there are still gaps in understanding how these models have been used to systematically support the evaluation and improvement of requirements quality. In this context, this study presents a systematic mapping aimed at investigating the application of LLMs in the evaluation, refinement, and improvement of textual requirements, considering aspects such as usage contexts, techniques adopted, tools employed, requirements quality metrics, and reported empirical evidence. The mapping was conducted based on the PICOC principle, using searches performed in four digital databases. The results indicate a predominance of the use of proprietary LLM models and suggest that these technologies have the potential to improve quality attributes such as clarity, completeness, and consistency. However, challenges persist regarding the reliability of the results and practical adoption, since most studies were conducted in academic contexts, with limited validation in real-world industrial settings.

1 INTRODUCTION

In recent decades, rapid advances in software and hardware technologies have consolidated systems as essential components in various sectors of society. Despite their apparent simplicity of use, software development is a complex process that requires a clear definition of the expected functionalities and an understanding of how these solutions can support the objectives of individuals and organizations (Arora et al., 2024). In this context, Requirements Engineering (RE) occupies a central position in software development, supporting the identification and documentation of stakeholder needs and ensuring alignment between the system to be developed and the client's expectations (Tasneem et al., 2025). Generally, these needs are expressed in natural language due to its accessibility and familiarity, especially in agile and innovation-oriented environments. However, this choice can make requirements more susceptible to problems such as ambiguity, incompleteness, and inconsistency, which, if not properly addressed, can negatively impact the development process and increase rework costs (Gentili and Falessi, 2025). Advances in Natural Language Processing (NLP) have revealed great potential for Requirements Engineering (RE), especially with the use of Large Language Models (LLMs). These models demonstrate a remarkable ability to understand and generate natural language texts, simulating human communication in an increasingly sophisticated manner. In this context, LLMs emerge as a promising resource for supporting requirements creation, whether for initial generation or for detecting and correcting quality issues (Arora et al., 2024).

The focus of this study is on organizing and summarizing the relationship between Large Language Models and the quality of software requirements written in natural language. Unlike more comprehensive reviews, this research focuses specifically on analyzing how these models can assist in detecting and resolving symptoms of poor quality in software requirements writing. Furthermore, through the investigation

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of analyses, evidence, and application contexts, the study seeks to understand how LLMs can contribute to improving requirements engineering activities in software projects, offering a more targeted perspective on the subject. This study contributes by (i) systematically characterizing LLM-based techniques for requirements quality improvement, (ii) mapping quality attributes and evaluation practices, and (iii) identifying gaps for industrial adoption.

2 BACKGROUND

2.1 Requirements Engineering and Software Requirements

Requirements Engineering (RE) is one of the most relevant stages in software development, as it directly influences the capabilities and quality of the final product (Veizaga et al., 2021). This process involves understanding and addressing multiple demands from different stakeholders, requiring a multidisciplinary approach and attention to the social interactions that permeate development. In practice, RE establishes the link between user needs and technical implementation, transforming expectations into clear, verifiable, and validatable requirements (Tasneem et al., 2025). Its main activities include elicitation, analysis, documentation, validation, and continuous management of requirements, conducted iteratively to allow adjustments and refinements throughout the software lifecycle (Udousoro, 2020).

Software requirements, in turn, describe the services the system must provide and the constraints on its operation, reflecting the client's needs in a given context of use. Due to the diversity of stakeholder perspectives and the need for a common language for communication, these requirements are generally specified in natural language (Gentili and Falessi, 2025). Furthermore, they can be classified as functional and non-functional: while the former define the behavior and functionalities of the system, the latter establish constraints and quality attributes applicable to the system as a whole.

2.2 Quality of Requirements

Assessing the quality of software requirements from the early stages of development is fundamental to reducing risks and costs associated with rework, especially when failures are only identified during implementation or after system delivery (Sommerville, 2011). To this end, quality metrics and standards have been proposed over time, most notably the ISO/IEC/IEEE 29148 standard (ISO, 2018), which consolidated and replaced previous standards focused on requirements engineering. This standard establishes processes, best practices, and guidelines for defining, validating, and managing requirements throughout the software lifecycle, as well as defining nine quality characteristics applicable to individual requirements, related to the system's capabilities, constraints, and quality attributes, as discussed in the Table 1.

The systematic adoption of these characteristics provides a consistent basis for guiding the writing and validation of requirements, especially when specified in natural language, contributing to increased reliability and comprehensibility of the developed software. In this context, the concept of Requirements Smells emerges, introduced by Femmer, which describes recurring symptoms of low quality in requirements written in natural language, capable of compromising attributes such as clarity, consistency, and verifiability (Femmer et al., 2017). Based on the ISO/IEC/IEEE 29148 guidelines (ISO, 2018), these symptoms signal potential weaknesses even in the specification phase.

The literature identifies a catalog of recurring Requirements Smells, including, among others, ambiguous or subjective language, unverifiable terms, comparatives and superlatives, incomplete references, excessive use of passive voice, and duplicated functionalities (Nascimento et al., 2021) (Rosadini et al., 2017). Although often subtle, these problems can generate divergent interpretations, information gaps, and redundancies, hindering the correct implementation of requirements. Thus, identifying and addressing Requirements Smells are relevant practices to support quality improvement in requirements engineering (Gentili and Falessi, 2025).

2.3 Large Language Model

Natural Language Processing (NLP) is a branch of Artificial Intelligence focused on the automated analysis and representation of language, encompassing both text and speech (Raiaan et al., 2024). In this context, Large Language Models (LLMs) stand out for using neural networks with billions of parameters, trained on large volumes of unlabeled data through self-supervised learning (Naveed et al., 2023). Among the best-known models is the GPT series from OpenAI (Radford et al., 2018), which are widely applied in natural language comprehension and generation tasks (Zhou et al., 2020).

One of the main ways of interacting with LLMs

Table 1: Quality attributes applicable to software requirements, as established by the ISO 29148 standard.

Attribute	Description
Appropriate	Presents an adequate level of abstraction, avoiding unnecessary constraints and implementation details.
Conforming	Follows a previously approved model or standard.
Complete	Covers all the needs of the system or software.
Correct	Is free of errors and accurately reflects the functionalities to be implemented in the software.
Unambiguous	Is clearly stated, understandable, and susceptible to only one interpretation.
Necessary	Defines an essential system aspect whose absence would compromise its operation.
Singular	Describes only one aspect of the system, without overlapping with other requirements.
Verifiable	Is formulated to prove or measure its fulfilment.
Feasible	Can be implemented considering the available resources, deadlines, and technologies.

is through prompt engineering, in which users design specific instructions to guide the generation of responses or the execution of tasks. This practice is common in evaluations and experiments involving LLMs. Furthermore, the models can be used in dialogue or question-answering modes, enabling natural language interactions (Liu et al., 2023) (Marques et al., 2024). Prompt design can be iteratively improved by testing different structures and instruction formats, following recognized patterns such as few-shot prompting, chain-of-thought prompting, and retrieval-augmented prompting (Velásquez-Henao et al., 2023).

3 RELATED WORK

(Necula et al., 2024) present a systematic review on the application of Natural Language Processing (NLP) and Artificial Intelligence techniques in Requirements Engineering, analyzing 309 publications from 1991 to 2023. The study identifies four phases of evolution in the field, ranging from approaches based on manual linguistic analysis to the more recent use of deep learning techniques. Among the main contributions are the automation of intensive tasks, the reduction of ambiguities in requirements, support for adaptability in distributed environments, and the bridging of theory and practice through empirical validations, highlighting the multidisciplinary nature of the field.

Complementarily, (Cheng et al., 2025) conducted a Systematic Literature Review on the use of Generative AI in Requirements Engineering, analyzing 238 studies published between 2019 and 2025. The work offers a comprehensive overview of the applications of generative models, predominantly GPT-based LLMs, across the ER phases, with a greater focus on requirements analysis and elicitation, while requirements management remains underexplored. The authors identified recurring and strongly interconnected challenges, especially reproducibility, hallucinations, and interpretability, which affected the reliability and practical adoption of these solutions. Furthermore, the review highlights that most proposals are still in early stages of development, with incipient industrial adoption and fragmented evaluation practices, emphasizing the need for methodological advancements, evaluation infrastructure, and standardization to enable the effective use of generative AI in Requirements Engineering.

Recent studies have broadly analyzed the use of NLP, LLMs, and AI in Requirements Engineering and Software Engineering, offering an overview of approaches, applications, and trends in the field. In contrast, this work focuses specifically on the relationship between LLMs and the quality of software requirements expressed in natural language, not addressing the quality of AI-based software or systems, unlike (Cheng et al., 2025) which adopts quality models geared towards AI systems, such as (ISO/IEC 25059, 2023). Furthermore, while studies such as (Hemmat et al., 2025) and (Cheng et al., 2025) analyze multiple phases of Requirements Engineering and various artifacts, including use cases, test cases, UML models, and formal requirements, this research deliberately restricts its scope to the specification of requirements in natural language. This approach, supported by well-defined research questions and selection criteria, allows for a more in-depth analysis of the quality characteristics of the requirements and of LLM-based strategies for their evaluation and improvement.

4 METHODOLOGY

The methodology adopted in this study followed the guidelines proposed by (Petersen et al., 2008). The methodology comprised three main stages: review planning (subsections 4.1 to 4.4), conducting the review (subsections 4.5 to 4.7), and analysis of the results (section 5). The research protocol was conducted with the support of the Parsifal tool[1], which enabled the structured recording of all stages. Data selection, extraction, and analysis were performed independently by two researchers, with consensus obtained through systematic comparison of results; any disagreements were resolved with the support of a third researcher. All data and artifacts from the review

[1]Access link: https://parsif.al

will be made publicly available in a GitHub repository[2], promoting transparency and replicability.

4.1 Research Questions

This work investigates the use of Large Language Models (LLMs) to support the improvement of software requirements quality, aiming to identify the techniques adopted, reported benefits, and associated challenges within requirements engineering. Based on this objective, the following research question was defined:

> *How have LLMs been used to support the improvement of software requirements quality?.*

From this question, eight research questions (RQs) were derived and organized into three groups. RQ1-RQ3 characterize the studies in terms of application domains, LLM-based tools, and prompt strategies. RQ4-RQ5 focus on how requirements quality is defined, evaluated, and addressed within requirements engineering activities. Finally, RQ6-RQ8 examine the scope and impact of the proposed approaches, empirical evidence, limitations and differences between human-in-the-loop and fully automated solutions.

Table 2: Research questions.

ID	Question
RQ1	What are the origins of the requirements used in the studies?
RQ2	What LLM-based tools, models or platforms are used?
RQ3	What prompt strategies and templates are employed?
RQ4	What quality attributes were explored?
RQ5	What metrics or evaluation methods are adopted?
RQ6	What empirical evidence supports improvements in requirements quality?
RQ7	What limitations, risks or challenges are reported?
RQ8	How are human participants involved in the evaluation and use of LLMs in studies?

4.2 Search Strategy and Terms

The search strategy was defined with the objective of identifying studies on the use of LLMs in Requirements Engineering, considering terminological variations associated with functional and non-functional requirements. To delimit the scope and guide the selection of terms, the PICOC strategy (Petticrew and Roberts, 2008) was adopted, applying only the Population and Intervention elements, due to the descriptive nature of this review, as presented in Table 3. Based on these terms, the following search string was defined, in order to minimize possible exclusion biases: *(”requirements engineering” OR ”functional requirements” OR ”non-functional requirements”) AND (”large language models” OR LLM OR LLMs).*

Table 3: PICOC Criteria.

PICOC Criteria	Description
Population	Requirements, Requirements Engineering, Functional Requirements, Non-functional requirements
Intervention	Large Language Models
Comparison	Not applicable
Outcome	Impacts on requirements production
Context	Software development

4.3 Source Selection

The selection of data sources in this study was based on two main criteria: (i) they had to be widely recognized and comprehensive scientific databases in the field of Software Engineering; and (ii) they needed to provide access to the full content of the articles, giving priority to open-access sources or to databases accessible through institutional or academic library agreements. Based on these criteria, the following databases were selected: ACM Digital Library, IEEE Xplore Digital Library, Web of Science, and Scopus. These sources are widely recognized for their relevance and broad coverage in the research area under investigation.

4.4 Inclusion and Exclusion Criteria

Inclusion and exclusion criteria were defined to ensure the relevance of the studies selected in relation to the research questions of this mapping. A study was considered valid only if it met all inclusion criteria and did not fall under any exclusion criteria.

Inclusion Criteria.

1. Studies addressing the use of LLMs within the context of Requirements Engineering in software development;
2. Publications written in English;
3. Studies focusing on activities of the Requirements Engineering process, such as elicitation, analysis, validation, or specification, with emphasis on the quality of requirements;
4. Studies that address interaction with LLMs through prompts;
5. Studies addressing requirements expressed in natural language.

[2]Access link: https://github.com/raianeeunice/article-files-iceis-2026

Exclusion Criteria.

1. Secondary or tertiary studies (systematic reviews, mappings, surveys, among others);
2. Works that do not present a clear relationship with the research questions;
3. Incomplete publications (extended abstracts, posters, presentations).

4.5 Study Selection

The study selection followed a sequential and systematic process. Initially, titles were analyzed to identify publications related to using LLMs in Requirements Engineering. Next, keywords and abstracts were reviewed to assess relevance, scope, methods, and evidence. The conclusions of the works were also examined to confirm their contribution to the topic. When these steps did not provide sufficient information, the full text was read to ensure that relevant studies were not prematurely excluded.

4.6 Quality Criteria

The quality assessment of the included studies aimed to ensure methodological robustness, reliability of evidence, and relevance of reported results (Yang et al., 2021). To this end, four quality assessment criteria (QC) were defined, presented in Table 4, based on the Kitchenham and Charters guidelines (Kitchenham et al., 2007). Each criterion was evaluated after a full reading of the studies, with a score of 1 assigned when fully met, 0.5 for partial meeting, and 0 for not met. Studies with a total score below 1.5 were excluded from the final analysis, even when thematically aligned with the research scope, in order to mitigate biases arising from low-quality evidence and strengthen the validity of the results (Yang et al., 2021).

Table 4: Quality Criteria.

ID	Quality Criterion
CQ1	Is the purpose of the study described clearly and objectively?
CQ2	Is the adopted methodology clearly presented, allowing for understanding and potential replication?
CQ3	Does the study discuss the benefits and limitations of using LLMs in requirements generation?
CQ4	Does the study present relevant contributions or perspectives for future research?

4.7 Execution

The initial database search resulted in the retrieval of 3,796 studies. After removing 1,212 duplicate records, 2,584 unique articles remained for analysis. Applying the inclusion and exclusion criteria in the screening stage eliminated 2,559 studies. Subsequently, a full reading of the remaining works was conducted, from which 10 primary studies fully met the defined criteria for review. The significant reduction in the initial set of studies highlights the comprehensive nature of the search strategy adopted, which retrieved works not aligned with the scope of this research, including studies focused on automatic requirements classification, formal specifications, UML modeling, and distinct domains of software development. In order to reduce the risk of omitting relevant studies not identified in the automatic search, the snowball search technique was systematically applied. From the 10 initially selected studies, the reference lists were independently analyzed by two researchers, resulting in the identification of 330 references and the selection of 10 candidate studies. These studies were subjected to the same inclusion and exclusion criteria, and any disagreements were resolved by consensus or, when necessary, with the support of a third researcher. As a result, 2 additional studies were included, totaling 12 primary studies. Finally, all selected studies were evaluated according to the quality criteria defined in Section 4.6, with no exclusions at this stage. The Table 5 presents the included studies, their IDs that will be used throughout the study, and their respective quality scores.

5 RESULTS AND DISCUSSION

The selected primary studies focus on a recent period, with publications between 2023 and 2025 and a clear predominance in 2024, highlighting the current relevance of the topic in the scientific literature. In general, the studies used requirements from fictitious scenarios or real public datasets. The predominance of academic and experimental contexts can largely be attributed to confidentiality restrictions that hinder the use and dissemination of requirements originating in industrial environments. Regarding the type of requirement analyzed, there is variation among the studies that explicitly provided this information. Most considered both functional and non-functional requirements, reflecting a concern with evaluating the applicability of LLMs in different dimensions of requirements quality. In studies that treated the types separately, the focus was predominantly on functional requirements.

Regarding the application of LLMs, their use is not limited to generating requirements from scratch. Several studies have explored these models to iden-

Table 5: Studies and their quality assessment score.

Reference	Study title	Study ID	QA score
(Biswas and Das, 2024)	ARIA-QA: AI-agent Based Requirements Inspection and Analysis through Question Answering	[S1]	4.0
(Zhang et al., 2024)	Defect Correction Method for Software Requirements Text Using Large Language Models	[S2]	4.0
(Görer and Aydemir, 2023)	Generating Requirements Elicitation Interview Scripts with Large Language Models	[S3]	4.0
(Görer and Aydemir, 2024)	GPT-Powered Elicitation Interview Script Generator for Requirements Engineering Training	[S4]	4.0
(Fantechi et al., 2023)	Inconsistency detection in natural language requirements using ChatGPT: a preliminary evaluation	[S5]	4.0
(Ronanki et al., 2023)	Investigating ChatGPT's Potential to Assist in Requirements Elicitation Processes	[S6]	3.0
(Alsaedi et al., 2025)	Investigating LLMs Potential in Software Requirements Evaluation	[S7]	4.0
(Jin et al., 2025)	iReDev: A knowledge-driven multi-agent framework for intelligent requirements development	[S8]	4.0
(Lubos et al., 2024)	Leveraging LLMs for the Quality Assurance of Software Requirements	[S9]	4.0
(Hasso et al., 2024)	Quest-RE: Question Generation and Exploration Strategy for Requirements Engineering	[S10]	3.0
(Krishna et al., 2024)	Using LLMs in Software Requirements Specifications: An Empirical Evaluation	[S11]	4.0
(Son, 2025)	Using Large Language Models in Software Requirements Analysis	[S12]	2.0

tify defects, detect quality problems, and produce refined versions of existing requirements. Regarding the phases of Requirements Engineering, LLMs have been employed in elicitation, for example in generating interview scripts and supporting communication with stakeholders; in specification, through the automatic production of documents such as Software Requirements Specification (SRS); and in analysis and validation, including the detection of redundancies, inconsistencies, ambiguities, and missing requirements, as well as improving completeness and textual clarity.

In general, the objectives associated with the use of LLMs focus on automating tasks that are intensive in human effort, improving the quality of requirements expressed in natural language, and supporting educational activities aimed at training requirements engineers. These findings reinforce the multifaceted nature of LLM applications, highlighting their potential to reduce manual effort, support systematic analyses, and improve the quality of requirements artifacts, albeit mostly in controlled contexts. A visual summary of the main information and characteristics of each study is presented in Figure 2 at the end of this section, to facilitate understanding and comparison of the results.

5.1 RQ1: What Are the Origins of the Requirements Used in the Studies?

The analyzed studies indicate three main origins for the evaluated requirements: own, external, and hybrid. The distribution among these categories was relatively balanced (5, 3, and 4 studies, respectively), although a slight preference for using requirements created by the authors themselves is observed. This choice, adopted in the works (S2, S4, S6, S8, S11), may indicate a gap in the literature, related to the absence or limited knowledge of widely accepted reference datasets for evaluating the quality of requirements, especially those containing explicit labels describing defects or quality problems of the requirement.

In the studies that adopted hybrid or external approaches, the PURE dataset stood out as the main source of requirements, being used in the works (S1, S9, S10). This dataset collects public requirements from various applications. However, the absence of labels indicating the quality of the requirements limits its use in supervised evaluations and hinders systematic comparisons across studies, potentially affecting the reproducibility and consistency of the reported empirical evidence. When PURE was not used, the authors relied on requirements from other studies (S3, S7, S12) or on alternative datasets (S5).

It is also observed that the business domains addressed are quite varied, both within the same study and across different works, demonstrating flexibility for evaluation across multiple contexts. Finally, there is a predominance of requirements written in English, with only one study (S2) identified that used requirements in another language, Chinese. This scenario suggests a preference for English, but also points to opportunities for future investigations with multilingual approaches, exploring more systematically the ability of LLMs to handle different languages.

5.2 RQ2: What LLM-Based Tools, Models or Platforms Are Used?

The studies predominantly used proprietary LLMs, particularly models from the GPT family, especially GPT-3.5 and GPT-4, generally accessed via ChatGPT and employed both in isolation and integrated into more complex architectures, such as custom agents and approaches based on Retrieval-Augmented Generation (RAG) (S1, S2, S4, S5, S6, S7, S9, S10, S11, S12). This concentration indicates a significant dependence on commercial models, whose updates and configurations are not fully controllable by researchers, which can affect the reproducibility of results over time.

To a lesser extent, some studies explored alternative LLMs, including open-source models such as BERT, Llama 2, and CodeLlama (S1, S8, S9, S10, S11), as well as other proprietary models such as Claude, Ernie BOT, and PaLM 2 (S1, S2, S6). Although these choices point to initiatives for diversification and comparison between architectures, they are still limited and do not allow for a systematic evaluation of the impact of model selection on results. Furthermore, the use of multiple LLMs in the same study is not always accompanied by clear descriptions of selection and parameterization criteria, which restricts comparability between approaches (S1, S2, S6, S9, S10, S12). This scenario highlights a gap in the literature and reinforces the need for more systematic investigations into the role of different LLMs, especially open-source models, in improving the quality of software requirements, focusing on reproducibility and methodological rigor.

5.3 RQ3: What Prompt Strategies and Templates Are Employed?

The analysis indicates the predominance of zero-shot prompting, generally employed in isolation or combined occasionally with other prompt engineering techniques. However, no systematic comparisons between different strategies were observed, nor were controlled evaluations investigating the relative impact of these techniques on the results obtained (S1, S5, S6, S7, S8, S9, 10, S11, 12). Nevertheless, some studies reported the use of additional approaches, such as prompt chaining, one-shot, role-based prompting (S1, S4), few-shot (S3), and chain-of-thought (S3). Only four studies combined different prompting techniques, not for comparative purposes, but to support the steps of their methodologies (S3, S6, S7, S8). However, it is observed that studies adopting AI agent-based architectures tend to employ more elaborate strategies, such as prompt chaining (S4) and role prompting (S8), aiming to decompose complex tasks into successive steps and guide the models to assume specific roles, aligned with guidelines and best practices in the field. This suggests an attempt to increase the coherence and suitability of responses to the context of requirements.

Regarding the structure of the prompts, the recurrent use of templates is observed, in which the requirements are provided as the main input, in line with zero-shot strategies. These templates tend to include contextualization of the requirement, task description, quality criteria, and definition of the response format. A specific case is study (S12), which analyzed a structured prompt pattern, called the Recipe Prompt Pattern, applied to the evaluation of requirements quality. Although some combinations of stimulus and interaction are associated, in the studies themselves, with more consistent results, the diversity of objectives and approaches limits the identification of a consolidated prompting pattern, even though recurring elements indicate promising configurations, such as structured prompts, individual requirements analysis, and prompt chaining.

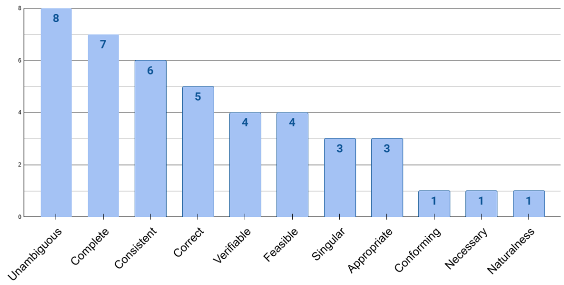

Figure 1: Distribution of quality attributes addressed in the selected studies.

5.4 RQ4: What Quality Attributes Were Explored?

The nine main attributes defined in ISO/IEC/IEEE 29148 were identified, as well as complementary attributes adopted by the twelve studies analyzed. Figure 1 shows the distribution of these attributes across the selected works. Most studies evaluated multiple quality attributes, with a predominance of the Unambiguous attribute, investigated in eight studies (S2, S3, S4, S6, S7, S8, S9, S11), and the Complete attribute, addressed in seven studies (S1, S2, S3, S4, S9, S10, S12). This result indicates that research efforts have focused on characteristics directly related to textual clarity and the identification of omissions, aspects more aligned with the linguistic capabilities of LLMs. In the case of completeness, the studies seek to verify whether the requirement adequately explains all the necessary aspects of the topic addressed, avoiding the absence of relevant information. In intermediate frequency, the attributes Consistent (a complementary attribute, considered in six studies: S1, S4, S5, S6, S7, S8), Correct (five studies: S3, S6, S7, S9, S11), Verifiable (four studies: S2, S8, S9, S11), and Feasible (four studies: S6, S7, S8, S9) stand out, reflecting a still limited, but growing, interest in attributes that require greater articulation between the content of the requirement and criteria of coherence, validation, or implementation.

In contrast, attributes such as Singular and Appropriate received limited attention, each being addressed in three studies (S6, S7, S9), while Conforming and Necessary were investigated sporadically, appearing in only one study (S9). This asymmetry suggests that attributes dependent on normative knowledge, a deeper understanding of the domain, or verification of adherence to previously approved models remain less explored, possibly because they require analyses that go beyond the purely textual evaluation of requirements. Among the complementary attributes, Naturalness was considered in only one study (S4), which evaluated the proximity of the language of the requirements to the communication style of a stakeholder, based on interview scripts generated by AI agents. The low recurrence of this attribute indicates that aspects related to communicative style and the adequacy of the requirements writing to human written language are still underexplored, pointing to opportunities for future investigations into the use of LLMs in improving requirements writing, focusing on reducing artificial constructs.

5.5 RQ5: What Metrics or Evaluation Methods Are Adopted?

In general, approaches can be organized into three groups: objective metrics, structured human evaluations, and, on a more focused basis, metrics specific to RAC-based architectures. Objective metrics encompass classic classification and similarity measures, such as accuracy, precision, recall, F1-score, BertScore, BLEU, TF-IDF (Term Frequency-Inverse Document Frequency), and Cohen's Kappa (S2, S5, S8, S9), as well as automatic text quality metrics, such as GRUEN (S4). These metrics were mainly used in studies that had predefined reference sets or annotations, allowing for more controlled comparisons between the outputs of the LLMs and criteria established by the authors themselves or by human evaluators.

In one specific study that adopted a RAG-based architecture (S1), metrics specific to the retrieval phase were employed, such as Context Recall, Context Precision, Faithfulness, and Answer Relevance, while the final evaluation of the generated content was based on the agreement between different LLM agents. In parallel, several studies resorted to human evaluations, some structured using judgment scales, such as the Likert scale, in which evaluators assigned scores. From 1 to 5, the documents generated by the LLMs were rated (S4, S11). Other studies adopted other techniques, such as error and satisfaction ratings with scores ranging from 0 to 10 (S2, S6). Finally, some studies were limited to a descriptive presentation of the results, without the explicit use of formal metrics or evaluation scales (S3, S7, S10, S12).

Although the diversity of metrics adopted among the studies makes comparability difficult, a relevant potential is observed in the combination of automatic metrics with structured human evaluations, as in (S4). Human evaluation, in particular, contributes to capturing perceptions of improvement more aligned with real software development contexts, indicating that hybrid approaches represent a promising direction for future research.

5.6 RQ6: What Empirical Evidence Supports Improvements in Requirements Quality?

The synthesis of empirical evidence indicates that improvements in requirements quality associated with the use of LLMs are primarily focused on textual and structural aspects of requirements expressed in natural language. The most recurrent findings point to the ability of these models to correct problems explicitly signaled in the prompts (S9), identify missing and redundant requirements (S12), and generate requirements with performance comparable to that of humans in attributes such as abstraction, atomicity, completeness, and explainability (S6). These results suggest a relevant role for LLMs in supporting the initial review and validation activities of requirements artifacts.

Additionally, it was observed that more informative and well-structured prompts are associated with more consistent improvements in the evaluated requirements (S2). Complementary evidence indicates that models such as GPT show greater consistency in the validation and correction of requirements when compared to other LLMs, such as CodeLlama (S11), in addition to contributing to the identification of incompleteness and inconsistency in automated analyses (S1). These findings also point to a potential reduction in manual effort resulting from the accelerated generation of requirements documents (S11). In specific contexts, such as the generation of interview scripts using GPT-4-based custom agents, the production of linguistically natural and coherent texts was observed (S4).

Despite these positive results, it is noted that such improvements are predominantly reported in experimental and controlled evaluations, with no evidence derived from prolonged use in industrial environments. Thus, although there is a consistent set of favorable indications, especially related to clarity, completeness, and efficiency, the empirical evidence still supports improvements mainly at a punctual and exploratory level, indicating the need for more robust studies that assess sustainable impacts on the practice of Requirements Engineering.

5.7 RQ7: What Limitations, Risks or Challenges Are Reported?

Despite the positive evidence, studies also reveal important limitations in the use of LLMs for improving requirements quality. A tendency is observed for some models, such as ChatGPT, to assign more favorable evaluations than those performed by human experts, raising questions about the reliability of automated analyses (S7). Limitations in ambiguity, feasibility, and inconsistency detection were also identified in some studies (S6). Additionally, weaknesses were observed in conducting interviews mediated by AI agents, especially regarding active listening. Although the agent follows the participants' responses, there is not always in-depth exploration of the topics based on these responses, leading to missed opportunities to explore aspects relevant to the construction and refinement of requirements (S4). Furthermore, there is evidence of inferior performance of LLMs compared to human experts in specific tasks, such as inconsistency detection, as well as difficulties in handling lengthy documents, resulting in degradation of metrics when the number of requirements increases (S5). These challenges are amplified by the fact that most assessments occur in controlled environments, with synthetic data or references created by the authors themselves and without the participation of real development teams, which reduces confidence in transferring these results to practice. Taken together, the findings indicate that the use of LLMs still involves substantial risks related to the reliability of the analyses, the validity of the assessments, and their suitability in real-world contexts, requiring caution in their adoption as an automated support for requirements quality.

5.8 RQ8: How Are Human Participants Involved in the Evaluation and Use of LLMs in Studies?

The analysis of the studies reveals significant heterogeneity in the composition of the human groups involved, including experienced Requirements Engineering specialists, students, and participants with initial experience in the field. This diversity reflects different experimental objectives, but also introduces relevant variations in the reliability and interpretation of the results. In general, human participation occurred in three main roles: (i) generation of requirements used as reference artifacts for comparison with LLM outputs; (ii) evaluation of the quality of the requirements produced by the models, frequently through structured judgments; and (iii) a combination of these two roles, in which participants acted both as producers and evaluators of the generated requirements. It is observed that most studies treated humans predominantly as sources of validation or baseline for comparison (S2, S4, S5, S6, S7, S9, S10, S11), reinforcing a view of LLMs as support tools, and not as active partners in the process. A relevant exception is study (S8), which explicitly explored human-AI agent collaboration. In this work (S8), AI agents iteratively perform multiple requirements development activities and interact with human experts to confirm, review, and refine the artifacts produced, indicating an initial move towards more integrated and interactive approaches. These findings suggest that, while human participation is central to validating results, its role is still largely instrumental, pointing to opportunities for future investigations that explore more integrated collaboration models between humans and tools, with greater adherence to real-world interaction and decision-making practices in Requirements Engineering.

5.9 Future Work

Based on the identified gaps, future research should prioritize empirical studies in real-world industrial environments, involving active development teams and projects in a commercial context. This direction is essential to assess the practical applicability of LLMs in improving requirements quality and understanding their impacts on workflows, stakeholder collaboration, and the productivity of Requirements Engineering teams.

Study	Prompting techniques	LLM model	RE phase	Type / interaction	Metrics
[S1]	Prompt chaining	Claude 3-opus (RAG), LLaMA-2, Mixtral 8x7B, and ChatGPT	Elicitation and analysis	Agent continuously generates prompts; example questions provided	Context Recall, Context Precision, Faithfulness, Answer Relevancy for RAG evaluation. No formal objective metric used for results
[S2]	Zero-shot	BERT, EmieBOT, GPT-3.5, and GPT-4	Analysis and validation	4 templates (incremental context variations); 1 requirement per call	Accuracy, Precision, Recall, F1, BLEU, Cosine Similarity, Cross-Entropy
[S3]	Few-shot + chain-of-thought	ChatGPT with GPT-3.5 and BARD	Elicitation	Batch vs. iterative; prompt templates	No formal objective metric used
[S4]	Prompt chaining	A customized GPT-4 agent (via RAG)	Elicitation	Chained prompts and multiple requests	TF-IDF, GRUEN, dispersion measures + human evaluation
[S5]	Zero-shot	ChatGPT with GPT3.5	Analysis and validation	Structured prompt for identifying inconsistent requirement pairs, with a predefined output format.	Precision and Recall
[S6]	Zero-shot and one-shot answers	ChatGPT	Elicitation	6 elicitation questions (domain: Trustworthy AI)	Measures of central tendency and dispersion
[S7]	Zero-shot and role method	ChatGPT-3.5, Claude 1, and Palm 2 via Poe Bot	Analysis and validation	Numerical prompt 0–10; 1 new chat per requirement; context with attribute definitions	No formal objective metric used
[S8]	Zero-shot, Role prompting and artifact-oriented prompting	GPT-4-turbo-2024-04-09	Elicitation, analysis, specification and validation	A combination of structured, role-oriented prompts linked by artifacts.	Convex hull volume (CHV), mean distance to centroid (MDC), precision, recall, F1 score, BLEU, BertScore and G-Eva.
[S9]	Zero-shot	Llama 2 (open-source) hosted on Replicate	Analysis	2 templates (binary evaluation + generation); 1 request per call	Cohen's Kappa
[S10]	Zero-shot	ChatGPT-4	Analysis and validation	Template generating tuple (requirement, topic, questions, objective)	No formal objective metric used
[S11]	Zero-shot	ChatGPT (GPT-4) and CodeLlama	Specification, analysis, and validation	Template for generation; template for evaluation; provide context/domains	Likert Scale (1–5) + comparison with human-generated documents
[S12]	Zero-shot	ChatGPT	Analysis	Adapted pattern (Recipe Prompt Pattern)	No formal objective metric used

Figure 2: Overview of the selected studies.

Another relevant aspect concerns human participation in the studies. Evidence indicates that human-LLM interaction has been largely restricted to traditional roles, such as requirements generation or evaluation, with only one study exploring human-agent collaboration in more depth. Thus, future investigations can advance the proposition and evaluation of more integrated collaborative models, in which humans and LLMs act in a complementary way throughout Requirements Engineering activities.

Furthermore, regarding requirements quality, future research can expand the set of quality attributes analyzed and advance the definition of more standardized and operationalizable metrics, in order to reduce the subjectivity of evaluations and favor comparability between studies. In this context, the creation of a public and standardized repository of requirements, encompassing different types, domains, defined quality errors, and quality levels, is a strategic direction to increase the reproducibility of experiments and contribute to the methodological maturity of research in the area.

Finally, it is recommended to broaden the scope of investigations to include requirements specified in languages other than english. The analysis of multilingual and multicultural scenarios can reveal limitations that are still largely unexplored in LLMs, since linguistic and contextual variations directly influence the clarity, precision, and interpretation of requirements. This approach contributes to the evaluation of the robustness, generalization, and adaptability of these models in contexts closer to the global practice of Requirements Engineering.

6 THREATS TO VALIDITY

Systematic literature reviews in the field of software engineering are subject to several threats to validity, which may arise throughout the planning, execution, and reporting stages of the studies (Zhou et al., 2016). Considering this limitation, the present systematic mapping followed the guidelines proposed by the authors, as the methodological process presents a similar cycle. Accordingly, the identified threats were organised into the domains of construct validity, external validity, internal validity, and conclusion validity. The main threats related to this study, as well as the strategies used to mitigate them, are presented below:

1. **Construct Validity:** To mitigate the risk of inadequate or incomplete search terms, the search sequence was pre-tested and adjusted before executing the protocol, in order to ensure greater relevance and comprehensiveness of the results. Nevertheless, it is possible that the adopted strat-

egy did not capture all relevant studies. Moreover, the formulated research questions may not fully cover all nuances of the investigated topic. To reduce this risk, the questions were refined through multiple iterations, based on preliminary literature readings, until clarity and alignment with the study's objectives were achieved among the authors. Even so, the possibility of omitting relevant works is acknowledged, mitigated by successive refinements and the application of the snowball sampling technique.

2. **External Validity:** The inclusion and exclusion criteria were defined based on the authors' knowledge of the topic, which may have led to some articles' inappropriate inclusion or exclusion. Two reviewers independently evaluated each study to reduce this risk, and they applied the selection criteria autonomously.
3. **Internal Validity:** The research was conducted in four widely recognised academic databases (ACM, IEEE Xplore, Scopus, and Web of Science). Although these represent a robust and comprehensive set, excluding other sources may have resulted in omitting potentially relevant studies.
4. **Conclusion Validity:** Data extraction followed a predefined protocol, including the strategy and collection format. The extraction form was developed in Parsifal to ensure consistency and alignment with the research questions. Extraction was performed by at least two researchers independently, with cross-checking to reduce bias and ensure the reliability of the results.

7 CONCLUSION

This work presents a systematic mapping of recent literature on the use of Large Language Models (LLMs) in supporting the improvement of the quality of requirements expressed in natural language. The results indicate that LLMs have been primarily employed in analysis, validation, specification, and elicitation activities, with evidence of potential for automating tasks intensive in human effort and supporting the identification and correction of quality problems, such as inconsistencies, ambiguities, and incompleteness. However, the findings also highlight relevant limitations, including the predominance of evaluations conducted in experimental contexts, the restricted attention to a limited subset of quality attributes, the heterogeneity of the metrics adopted, and the absence of standardized requirements databases for quality evaluation, with labels that explicitly characterize the types of failures. Furthermore, there is a still incipient exploration of collaborative approaches between humans and AI agents that systematically incorporate domain knowledge. Taken together, the results indicate that, while promising, the uses of LLMs for improving requirements quality still demand more robust, methodologically standardized investigations aligned with real-world Requirements Engineering practices so that their benefits can be consolidated and generalized.

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